

Digital Transformation, Digitization, and Digital Modernization: An ROI-Oriented Decision Framework for Enterprise Investment

Bhagyeshkumar Chokhawala¹[0009-0001-5146-832X]

¹Comcast, Philadelphia, Pennsylvania, USA

bhagyeshkumar_chokhawala@comcast.com

Abstract. Digital initiatives are frequently labeled as transformations even when their primary purpose is converting analog information, renewing legacy technology, or optimizing an existing process. This conceptual paper distinguishes digitization, digital modernization, and digital transformation and proposes an ROI-oriented decision framework for selecting, governing, and evaluating enterprise investments. Building on an original practitioner article and an integrative synthesis of digital-transformation, technology-adoption, resilience, and benefits-realization literature, the paper positions digitization as information conversion and workflow enablement, digital modernization as architectural and platform renewal, and digital transformation as material change to value creation, customer experience, processes, capabilities, or operating models. The proposed framework links strategic intent, current- and future-state analysis, capability dependencies, total lifecycle cost, adoption-adjusted benefits, scenario-based ROI, resilience, and post-implementation measurement. It extends conventional ROI by separating anticipated from actual returns and by incorporating adoption, data quality, change costs, downtime, risk reduction, and option value. An illustrative enterprise case demonstrates that a lower-cost digitization initiative can yield rapid payback, while modernization and transformation may create larger long-term value and strategic flexibility. The framework helps leaders avoid category confusion, underfunded change management, and feature-centric business cases that fail to measure realized outcomes.

Keywords: digital transformation; digitization; digital modernization; return on investment; benefits realization; enterprise architecture; technology adoption; resilience

1. Introduction

Digital transformation has become a broad label for almost any technology-enabled initiative. Yet replacing paper with electronic records, replatforming a legacy application, and redesigning an enterprise business model are not equivalent interventions. They differ in strategic intent, scope, risk, capability requirements, time-to-value, and the type of return to measure. The original practitioner article underlying this paper argued that organizations should first clarify whether they need digitization, modernization, or transformation, and then test the initiative through current- and future-state analyses and ROI scenarios [1].

The academic literature similarly distinguishes digitization, digitalization, and digital transformation, while emphasizing that transformation involves significant organizational change rather than technology deployment alone [2]-[5]. However, practitioners continue to face two recurring problems. First, initiatives are frequently misclassified, leading to the wrong governance model and success measures being applied. Second, ROI is often calculated as a static pre-approval number while adoption, transition costs, business continuity, and actual post-implementation benefits receive insufficient attention.

This paper addresses those problems through three research questions: (RQ1) How can organizations distinguish digitization, digital modernization, and digital transformation according to dominant strategic intent? (RQ2) Which lifecycle costs, benefit drivers, and adoption factors should be included in ROI assessment? (RQ3) How can scenario analysis and post-implementation measurement improve investment governance?

The contribution is a practical conceptual framework that integrates initiative classification, capability and change dependencies, scenario-based financial evaluation, resilience, and benefits realization. The framework is not presented as an empirically validated causal model. Instead, it is a design-oriented synthesis intended for subsequent case-study, survey, and longitudinal validation.

2. Conceptual Background

2.1 Digitization

Digitization is the conversion of information or activity from analog or manually handled forms into machine-readable digital forms. Examples include scanning documents, replacing paper forms with electronic ones, capturing signatures digitally, or converting manual records into structured data. Digitization can improve accessibility, searchability, data capture, and workflow speed, but it does not necessarily alter the underlying value proposition, organizational design, or end-to-end process. Literature often treats digitization as the first stage in a broader progression toward digitalization and transformation [3]-[5].

2.2 Digital Modernization

Digital modernization renews the technological foundation that supports business capabilities. Typical interventions include cloud migration, application re-architecture, API enablement, data-platform renewal, user-interface replacement, automation, security uplift, and retirement of unsupported technologies. Modernization may substantially change architecture and delivery practices while leaving the core product, customer promise, or operating model relatively stable. Its value therefore includes not only direct cost reduction, but also reliability, scalability, maintainability, risk reduction, and strategic option value. Modernization often enables later transformation, but it should not be mistaken for transformation when the business and operating models remain largely unchanged.

2.3 Digital Transformation

Digital transformation is a broader process of organizational renewal enabled by digital technologies. Reviews describe it as significant change to an entity through combinations of information, computing, communication, and connectivity technologies [2], and as a phenomenon affecting customer behavior, business models, organizational structures, capabilities, and performance metrics [3]. Unified-definition research emphasizes the need to distinguish transformation from adjacent concepts such as digitization and digitalization [4]. Transformation therefore requires coordinated changes in value creation, technology use, organizational structure, and financial logic [6], supported by dynamic capabilities for sensing opportunities, seizing them, and reconfiguring resources [8].

2.4 Why the Distinction Matters

Classification affects decision rights and measurement. A digitization initiative may be governed as a process-improvement investment with short-cycle productivity metrics. Modernization requires architecture, migration, operational-risk, and technical-debt measures. Transformation requires executive sponsorship, cross-functional ownership, new capabilities, redesigned customer or employee journeys, and outcome metrics that may take several years to mature. Treating all three as transformation can inflate expectations; treating transformation as a technology upgrade can underfund organizational change.

Dimension	Digitization	Digital modernization	Digital transformation
Dominant intent	Convert information or manual activity into digital form	Renew legacy platforms, architecture, data, security, and delivery foundations	Change value creation, experience, process, capabilities, or operating model
Typical scope	Task, record, document, or bounded workflow	Application, platform, integration, infrastructure, or technology portfolio	End-to-end journey, product, ecosystem, business unit, or enterprise
Business change	Low to moderate	Moderate; often process-preserving	High; may redefine roles, decisions, products, and incentives
Technology change	Low to moderate	High	Moderate to high, selected to enable strategic change
ROI horizon	Short	Medium	Medium to long
Primary measures	Cycle time, error reduction, labor avoided, accessibility	TCO, reliability, release speed, risk reduction, scalability, technical-debt retirement	Revenue, retention, experience, market responsiveness, operating-model productivity, strategic options
Common failure	Automating a weak process without redesign	Replatforming without measurable business value or adoption	Technology-first program lacking ownership, capability change, and benefit realization

Table 1. Comparison of digitization, digital modernization, and digital transformation.

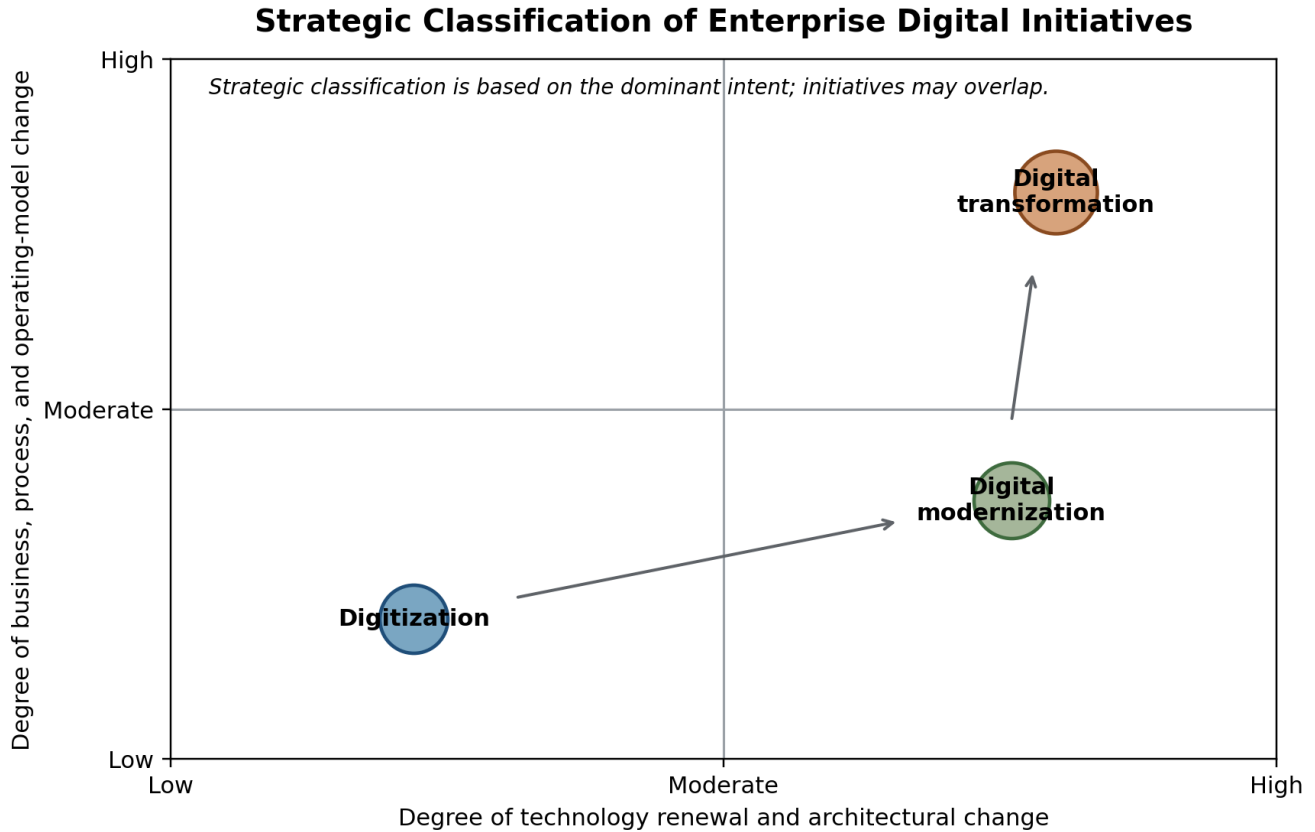


Figure 1. Strategic classification by dominant technological and organizational change.

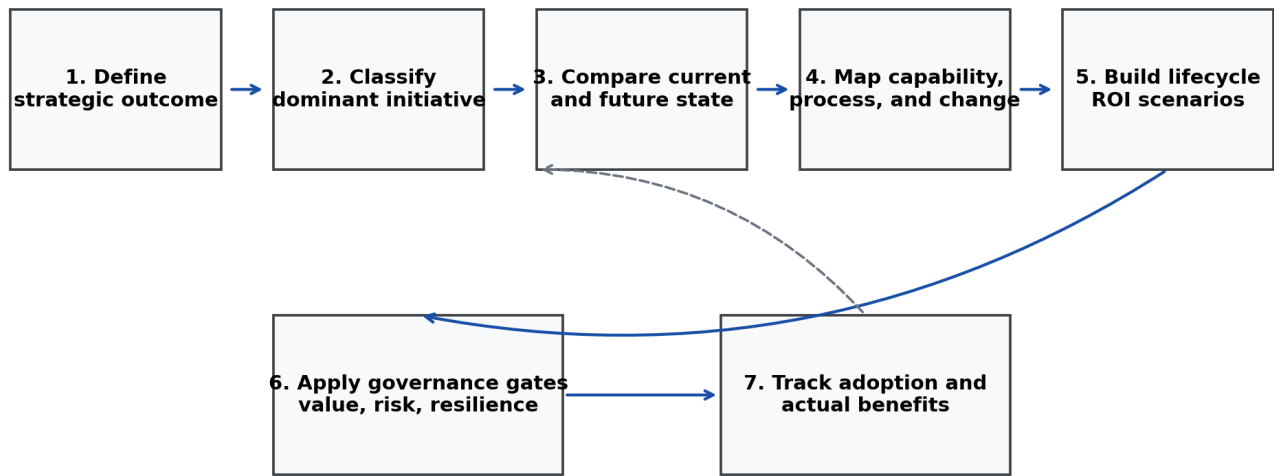
3. Method of Conceptual Development

The paper uses a design-oriented conceptual synthesis. The starting point was the practitioner framework published by Chokhawala [1], which proposed differentiating digital initiatives, conducting current- and future-state SWOT analysis, assessing value propositions, and calculating time-based, optimistic/pessimistic, and anticipated/actual ROI. These ideas were refined against peer-reviewed literature on digital transformation [2]-[8], benefits realization [9]-[12], user adoption [13], information-system ROI [14], and operational resilience [15]-[16].

The synthesis followed four steps. First, constructs were separated according to dominant intent and scope. Second, implementation dependencies were organized into business, process, people, data, technology, and operational-resilience categories. Third, a lifecycle ROI model was developed to include transition and realization factors that are commonly omitted from static business cases. Fourth, governance stages were arranged into a feedback loop so that actual outcomes can change assumptions, priorities, and funding.

4. ROI-Oriented Digital Investment Decision Framework

ROI-Oriented Digital Investment Decision Framework



Continuous feedback: actual outcomes update assumptions, sequencing, and funding decisions.

Figure 2. Proposed ROI-oriented digital investment decision framework.

4.1 Define the Strategic Outcome

The process begins with a measurable future state rather than a predetermined technology. The outcome should describe the stakeholder, the required change, the baseline, the target, and the decision horizon. Examples include reducing customer onboarding time from five days to one hour, reducing critical-platform recovery time from eight hours to thirty minutes, or creating a digital channel that contributes a specified share of new revenue.

4.2 Classify the Dominant Initiative

The initiative is classified according to its dominant purpose. A program may include all three categories, but one should lead. For example, a transformation of customer onboarding may require modernization of identity, API, and data platforms and digitization of documents. The business case should distinguish enabling investments from the primary outcome to prevent infrastructure costs from being justified by vague transformation language.

4.3 Compare Current and Future States

A structured current/future-state assessment identifies strengths, weaknesses, opportunities, threats, capability gaps, constraints, and assumptions. The analysis should be repeated after the proposed design is defined. Risks that disappear

should be traceable to a funded intervention; risks that remain should be accepted, mitigated, transferred, or monitored. This converts SWOT from a descriptive workshop into an investment-control mechanism.

4.4 Map Capability, Process, and Change Dependencies

Benefits are not produced by software alone. The framework maps required business capabilities, process changes, data quality, roles, policies, skills, incentives, operating procedures, architecture, security controls, support models, and adoption activities. This is consistent with benefits-realization research, which links IT enablers to business changes and accountable benefit owners [9]-[12].

4.5 Build Lifecycle ROI Scenarios

The business case should include pessimistic, base, and optimistic scenarios, each with explicit assumptions about adoption, benefit ramp, implementation cost, delay, operational disruption, and residual risk. Time-based analysis prevents a large upfront investment from being judged only by early negative returns and prevents distant benefits from being accepted without discounting or probability adjustment.

4.6 Apply Governance Gates

Funding decisions should use gates for strategic alignment, architecture, security, resilience, data readiness, change readiness, economic value, and benefit ownership. A project can pass delivery milestones while still failing to realize value. Governance must therefore continue beyond deployment.

4.7 Track Adoption and Actual Benefits

After release, anticipated ROI is replaced with actual cost and benefit evidence. Adoption, utilization, customer behavior, operational performance, incident rates, and process conformance are reviewed against baseline and target. Variance should trigger corrective action, benefit reforecasting, scope adjustment, or termination when the investment no longer supports strategic outcomes.

5. Lifecycle and Adoption-Adjusted ROI Model

A single ROI percentage is insufficient for complex digital investments. ROI should be used with net present value (NPV), payback period, total cost of ownership, risk exposure, and nonfinancial outcome measures. ROI remains useful when its boundary, horizon, assumptions, and uncertainty are explicit [14].

5.1 Total Lifecycle Cost

The cost base should extend beyond acquisition and development. It should include discovery, architecture, software, infrastructure, integration, migration, data remediation, cybersecurity, training, communications, dual running, downtime, productivity loss, vendor transition, operating support, decommissioning, compliance, and contingency.

$$TLC = C_{acquisition} + C_{build} + C_{migration} + C_{change} + C_{downtime} + C_{run} + C_{risk}$$

5.2 Benefit Categories

Benefits should be separated into revenue growth, cost reduction, productivity, working-capital improvement, experience improvement, quality, risk reduction, resilience, compliance, learning, and strategic option value. Intangible benefits should be connected to observable proxies rather than inserted as unsupported financial values. For example, improved customer experience can be linked to conversion, retention, reduced contact rate, or service recovery cost.

5.3 Adoption-Adjusted Benefits

A technically available capability does not automatically produce value. User acceptance research shows that performance expectancy, effort expectancy, social influence, and facilitating conditions influence adoption [13]. The paper therefore proposes an adoption-adjusted benefit model:

$$Realized\ Benefit_t = Potential\ Benefit_t \times Adoption_t \times Effective\ Use_t \times Process\ Conformance_t \times Data\ Quality_t$$

Each multiplier ranges from zero to one. The equation prevents a business case from assuming full benefit on the deployment date and makes training, experience design, process redesign, and data quality economically visible.

5.4 Discounted ROI and NPV

$$NPV = Sum[(Benefit_t - Cost_t) / (1 + r)^t] - Initial\ Investment$$

$$ROI_H = (PV(Benefits_H) - PV(Costs_H)) / PV(Costs_H) \times 100$$

The horizon H should match the useful decision period, and the discount rate r should reflect organizational policy and risk. Modernization may produce option value by enabling faster future changes; such value should be documented separately or estimated using decision-tree or real-options techniques rather than hidden in an inflated benefit estimate.

5.5 Scenario and Variance Analysis

The framework requires pessimistic, base, and optimistic cases. Pessimistic estimates use lower benefit realization, slower adoption, higher costs, and longer delays. Optimistic estimates use higher but defensible benefits and lower transition friction. Expected value can be calculated by probability weighting, while sensitivity analysis identifies assumptions that most influence the decision.

$$Expected\ NPV = p_{pess} NPV_{pess} + p_{base} NPV_{base} + p_{opt} NPV_{opt}$$

Anticipated ROI is calculated during approval; actual ROI is recalculated with observed data. The variance between them should be explained by cost, schedule, adoption, demand, quality, operational, and external factors. This closes the learning loop and improves future estimates.

Factor	Example measures	Common omission risk
Strategic alignment	Contribution to outcome, capability, regulatory or market objective	Funding technology with no accountable business result
Acquisition and build	Licenses, engineering, integration, testing, architecture	Understating cross-system and nonfunctional work
Migration and transition	Data cleansing, conversion, dual run, cutover, decommissioning	Ignoring coexistence and retirement cost
Adoption and change	Training, communications, role redesign, incentives, support	Assuming deployment equals utilization

Operational disruption	Downtime, productivity dip, error recovery, backlog	Overstating early benefits
Run and support	Cloud consumption, operations, vendors, security, enhancements	Comparing capital cost without lifecycle TCO
Benefits	Revenue, cost, productivity, quality, experience, risk reduction	Double counting or treating activity as benefit
Resilience and continuity	Availability, recovery time, recovery point, degraded-mode capability	Valuing efficiency while ignoring outage exposure
Actual realization	Adoption, utilization, baseline variance, owner sign-off	Ending governance at go-live

Table 2. ROI factors and typical measurement gaps.

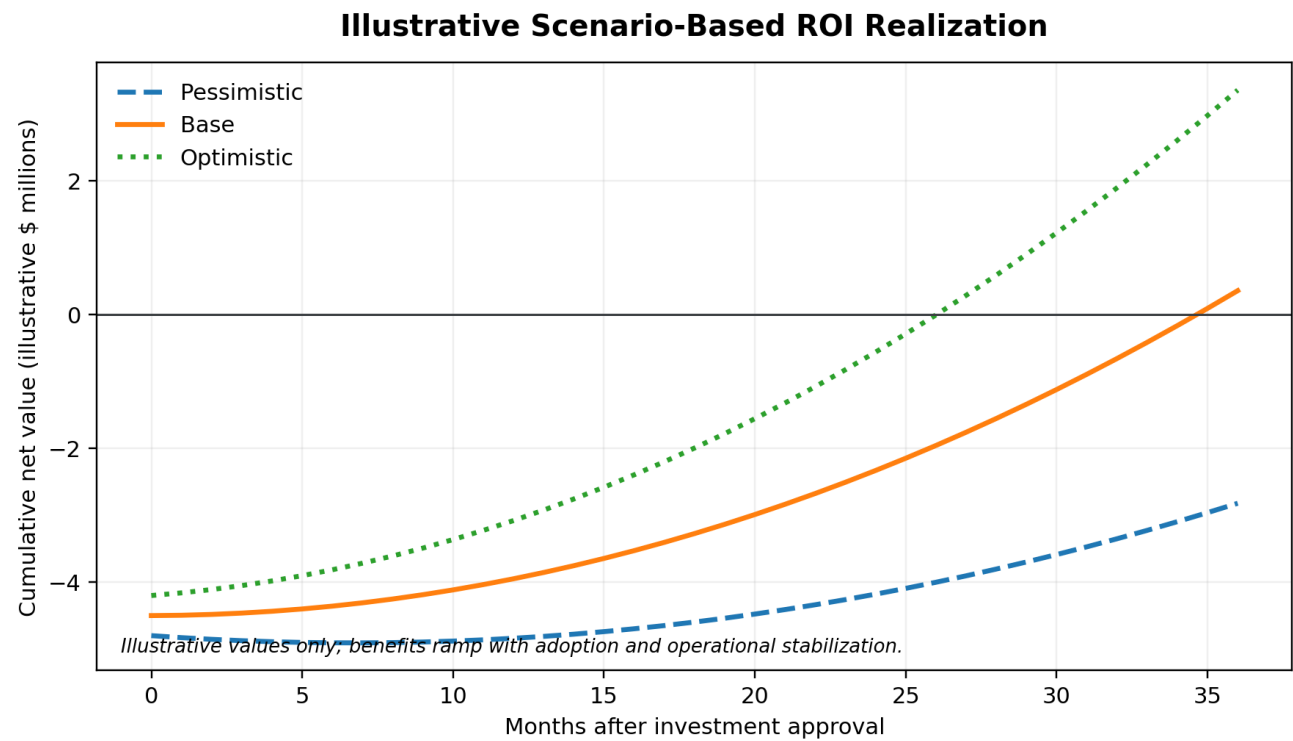


Figure 3. Illustrative pessimistic, base, and optimistic cumulative net-value trajectories.

6. Illustrative Enterprise Application

Consider an enterprise seeking to improve customer onboarding. Three alternatives are assessed over four years at an 8% discount rate. The example is hypothetical and intended to demonstrate classification and measurement, not to report empirical findings.

Alternative	Dominant scope	Upfront cost	Four-year benefits	PV	Four-year PV costs	NPV	Discounted ROI
-------------	----------------	--------------	--------------------	----	--------------------	-----	----------------

A. Digitization	Electronic forms, signatures, and document capture	\$0.40M	\$0.95M	\$0.67M	\$0.29M	42.9%
B. Modernization	Replatform onboarding services, APIs, data, and security	\$2.20M	\$3.65M	\$3.03M	\$0.62M	20.6%
C. Transformation	Redesign journey, decisioning, channels, roles, and partner integration	\$4.50M	\$9.95M	\$6.39M	\$3.57M	55.8%

Table 3. Hypothetical four-year financial comparison of three onboarding alternatives.

Alternative A offers a bounded, relatively rapid improvement and may be the correct decision when the strategic need is document conversion and workflow speed. Alternative B has a lower standalone discounted ROI in the example, but it may be required to reduce operational risk and enable future products. Alternative C produces the largest NPV because it changes the end-to-end experience and operating model, but it also depends most heavily on adoption, process redesign, governance, and cross-functional execution. Selecting the highest percentage ROI without considering NPV, risk, sequencing, and strategic dependencies would therefore be misleading.

A portfolio decision may sequence the alternatives rather than select only one. Digitization can create early benefits and data visibility; modernization can remove architectural constraints; transformation can then exploit the new capabilities. Each layer should retain its own benefit owner and metrics so that enabling work is not hidden inside an undifferentiated program.

7. Discussion

7.1 The Dominant-Intent Principle

The framework proposes that labels be assigned according to dominant strategic intent, not marketing language or technology novelty. This principle accepts overlap while preserving accountability. A cloud migration remains modernization unless it materially changes value creation or the operating model. An electronic form remains digitization unless the surrounding process, decisions, and stakeholder experience are redesigned. Conversely, a transformation program can include conventional technologies if they enable a material strategic change.

7.2 ROI as a Managed Hypothesis

ROI should be treated as a hypothesis that evolves from anticipated to actual evidence. Benefits-realization research emphasizes that value requires management beyond project delivery [9]-[12]. The proposed model operationalizes this by assigning benefit ownership, adoption factors, baselines, scenario assumptions, and post-implementation reviews. The objective is not to create an artificially precise forecast, but to expose uncertainty and make assumptions governable.

7.3 Resilience and Continuity as Economic Factors

Digital programs can temporarily increase fragility through migrations, new dependencies, and complex integrations. NIST guidance treats continuity and cyber resilience as lifecycle concerns supporting mission and business functions [15]-[16]. Downtime, degraded operation, recovery capability, and external-provider dependencies should therefore be included in business cases. Modernization investments that reduce outage probability or recovery time may have substantial value even when direct labor savings are limited.

7.4 Implications for Enterprise Architecture and Governance

Enterprise architecture can use the classification matrix to connect business outcomes, capabilities, process changes, data, applications, platforms, and transition states. Portfolio governance can then compare initiatives using category-appropriate metrics while preserving a common lifecycle value model. Transformation funding should include organizational change and benefit realization; modernization funding should include decommissioning and operational transition; digitization funding should test whether the underlying process should first be simplified or redesigned.

8. Design Propositions for Future Validation

- P1: Digital initiatives classified by dominant intent will show higher alignment between governance measures and expected outcomes than initiatives classified only by technology type.
- P2: Business cases that include adoption-adjusted benefits will produce smaller anticipated-to-actual ROI variance than business cases that assume immediate full utilization.
- P3: Explicit inclusion of migration, dual-running, downtime, and decommissioning costs will improve total lifecycle cost accuracy for modernization programs.
- P4: Transformation initiatives with named benefit owners and post-implementation measurement will realize a higher proportion of planned benefits than initiatives governed only through delivery milestones.
- P5: Scenario-based ROI and sensitivity analysis will improve termination, sequencing, and scope decisions compared with single-point ROI estimates.

9. Limitations and Future Research

This paper is conceptual and uses a hypothetical case. The categories are intentionally simplified and may require adaptation by industry, regulatory environment, investment scale, and organizational maturity. Modernization and transformation are particularly difficult to separate when platform renewal directly enables a new digital product. Financial estimates may also omit externalities, ecosystem effects, or option value that cannot be measured reliably at approval time.

Future research should validate the framework through comparative case studies and longitudinal analysis of anticipated and actual ROI. Survey research could test whether classification clarity, benefit ownership, adoption measurement, and resilience planning predict benefit realization. Additional work should develop industry-specific weighting models and investigate how AI-enabled initiatives alter the distinction between modernization and transformation.

10. Conclusion

Digitization, digital modernization, and digital transformation are related but strategically different forms of enterprise change. Digitization converts information and manual activity into digital form. Modernization renews the

technological foundation. Transformation changes how the organization creates value, serves stakeholders, operates, and competes. Misclassification leads to mismatched expectations, incomplete funding, and weak measurement.

The proposed framework begins with a measurable future state, identifies the dominant initiative type, compares current and future conditions, maps capability and change dependencies, evaluates lifecycle ROI under multiple scenarios, applies value and resilience gates, and measures actual benefits after deployment. Its central message is that digital investment success should be judged not by the number of features delivered or technologies adopted, but by sustained, measurable business outcomes relative to the full cost and risk of change.

References

1. G. Vial, "Understanding digital transformation: A review and a research agenda," *Journal of Strategic Information Systems*, vol. 28, no. 2, pp. 118-144, 2019, doi: 10.1016/j.jsis.2019.01.003.
2. P. C. Verhoef et al., "Digital transformation: A multidisciplinary reflection and research agenda," *Journal of Business Research*, vol. 122, pp. 889-901, 2021, doi: 10.1016/j.jbusres.2019.09.022.
3. C. Gong and V. Ribiere, "Developing a unified definition of digital transformation," *Technovation*, vol. 102, art. 102217, 2021, doi: 10.1016/j.technovation.2020.102217.
4. C. Legner et al., "Digitalization: Opportunity and challenge for the business and information systems engineering community," *Business & Information Systems Engineering*, vol. 59, pp. 301-308, 2017, doi: 10.1007/s12599-017-0484-2.
5. C. Matt, T. Hess, and A. Benlian, "Digital transformation strategies," *Business & Information Systems Engineering*, vol. 57, pp. 339-343, 2015, doi: 10.1007/s12599-015-0401-5.
6. T. Hess, C. Matt, A. Benlian, and F. Wiesböck, "Options for formulating a digital transformation strategy," *MIS Quarterly Executive*, vol. 15, no. 2, pp. 123-139, 2016.
7. K. S. R. Warner and M. Wäger, "Building dynamic capabilities for digital transformation: An ongoing process of strategic renewal," *Long Range Planning*, vol. 52, no. 3, pp. 326-349, 2019, doi: 10.1016/j.lrp.2018.12.001.
8. J. Peppard, J. Ward, and E. Daniel, "Managing the realization of business benefits from IT investments," *MIS Quarterly Executive*, vol. 6, no. 1, pp. 1-11, 2007.
9. C. E. M. Serra and M. Kunc, "Benefits Realisation Management and its influence on project success and on the execution of business strategies," *International Journal of Project Management*, vol. 33, no. 1, pp. 53-66, 2015, doi: 10.1016/j.ijproman.2014.03.011.
10. A. Badewi, "The impact of project management and benefits management practices on project success: Towards developing a project benefits governance framework," *International Journal of Project Management*, vol. 34, no. 4, pp. 761-778, 2016, doi: 10.1016/j.ijproman.2015.05.005.
11. Project Management Institute, *Benefits Realization Management: A Practice Guide*. Newtown Square, PA, USA: Project Management Institute, 2018.
12. V. Venkatesh, M. G. Morris, G. B. Davis, and F. D. Davis, "User acceptance of information technology: Toward a unified view," *MIS Quarterly*, vol. 27, no. 3, pp. 425-478, 2003, doi: 10.2307/30036540.
13. A. Botchkarev and P. Andru, "A return on investment as a metric for evaluating information systems: Taxonomy and application," *Interdisciplinary Journal of Information, Knowledge, and Management*, vol. 6, pp. 245-269, 2011, doi: 10.28945/1535.
14. M. Swanson et al., *Contingency Planning Guide for Federal Information Systems*, NIST Special Publication 800-34 Rev. 1. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2010, doi: 10.6028/NIST.SP.800-34r1.
15. R. Ross et al., *Developing Cyber-Resilient Systems: A Systems Security Engineering Approach*, NIST Special Publication 800-160, vol. 2, rev. 1. Gaithersburg, MD, USA: National Institute of Standards and Technology, 2021, doi: 10.6028/NIST.SP.800-160v2r1.